

Matrix Equations

Q1.

A system of three equations is defined by

$$\begin{aligned} kx + 3y - z &= 3 \\ 3x - y + z &= -k \\ -16x - ky - kz &= k \end{aligned}$$

where k is a positive constant.

Given that there is no unique solution to all three equations,

(a) show that $k = 2$

(2)

Using $k = 2$

(b) determine whether the three equations are consistent, justifying your answer.

(3)

(c) Interpret the answer to part (b) geometrically.

(1)

(Total for question = 6 marks)

Q2.

The population of chimpanzees in a particular country consists of juveniles and adults. Juvenile chimpanzees do not reproduce.

In a study, the numbers of juvenile and adult chimpanzees were estimated at the start of each year. A model for the population satisfies the matrix system

$$\begin{pmatrix} J_{n+1} \\ A_{n+1} \end{pmatrix} = \begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix} \begin{pmatrix} J_n \\ A_n \end{pmatrix} \quad n = 0, 1, 2, \dots$$

where a is a constant, and J_n and A_n are the respective numbers of juvenile and adult chimpanzees n years after the start of the study.

(a) Interpret the meaning of the constant a in the context of the model.

(1)

At the start of the study, the total number of chimpanzees in the country was estimated to be 64 000

According to the model, after one year the number of juvenile chimpanzees is 15 360 and the number of adult chimpanzees is 43 008

(b) (i) Find, in terms of a

$$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1}$$

(3)

(ii) Hence, or otherwise, find the value of a .

(3)

(iii) Calculate the change in the number of juvenile chimpanzees in the first year of the study, according to this model.

(2)

Given that the number of juvenile chimpanzees is known to be in decline in the country,

(c) comment on the short-term suitability of this model.

(1)

A study of the population revealed that adult chimpanzees stop reproducing at the age of 40 years.

(d) Refine the matrix system for the model to reflect this information, giving a reason for your answer.

(There is no need to estimate any unknown values for the refined model, but any known values should be made clear.)

(2)

(Total for question = 12 marks)

Q3.

Tyler invested a total of £5000 across three different accounts; a savings account, a property bond account and a share dealing account.

Tyler invested £400 more in the property bond account than in the savings account.

After one year

- the savings account had increased in value by 1.5%
- the property bond account had increased in value by 3.5%
- the share dealing account had **decreased** in value by 2.5%
- the total value across Tyler's three accounts had increased by £79

Form and solve a matrix equation to find out how much money was invested by Tyler in each account.

(7)

(Total for question = 7 marks)

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$\begin{vmatrix} k & 3 & -1 \\ 3 & -1 & 1 \\ -16 & -k & -k \end{vmatrix} = k(k+k) - 3(-3k+16) - 1(-3k-16)$	M1	2.1
	Solves $\det = 0 \Rightarrow 2k^2 + 12k - 32 = 0$ or $k^2 + 6k - 16 = 0$ To achieve $k = 2$ ($k = -8$ must be rejected)	A1	1.1b
		(2)	
	Special case $\begin{vmatrix} 2 & 3 & -1 \\ 3 & -1 & 1 \\ -16 & -2 & -2 \end{vmatrix} = 2(2+2) - 3(-3 \times 2 + 16) - 1(-3 \times 2 - 16)$ Shows $\det = 0$, therefore when $k = 2$ there is no unique solution	M1 A0	2.1 1.1b

(b)	Eliminates z to achieve two equations in x and y e.g. $5x + 2y = 1$ $-10x - 4y = -2$ $20x + 8y = 4$	Eliminates x to achieve two equations in y and z e.g. $11y - 5z = 13$ $22y - 10z = 26$ $-22y - 10z = -26$	Eliminates y to achieve two equations in x and z e.g. $11x + 2z = -3$ $22x + 4z = -6$ $-44x - 8z = 12$	M1 A1	3.1a 1.1b
	Must give a reason : e.g. Two equations are a linear multiple of each other e.g. shows they are the same equation therefore the equations are consistent .			A1	2.4
	Alternative Eliminates two different variables to form two equations, should be one equation from two of the three sections in the main scheme. e.g. $5x + 2y = 1$ and $11y - 5z = 13$ rearranges and substitutes in to one of the original equations in three variables. e.g. $2x + 3\left(\frac{1 - 5x}{2}\right) - \left(\frac{-3 - 11x}{2}\right) = 3$			M1	3.1a
	Correct equations e.g. $5x + 2y = 1$ and $11y - 5z = 13$			A1	1.1b
	Shows that the equations are a solution e.g. $3 = 3$ therefore consistent			A1	2.4
(c)	The three planes form a sheaf .			B1	2.2a
				(1)	
(6 marks)					

Notes:
(a) M1: Finds the determinant of the matrix corresponding to the system of equations. A1: Sets determinant = 0 and solves their 3TQ to achieve $k = 2$ ($k = -8$ must be rejected)
(a) Special case M1A0: Uses $k = 2$ and finds the determinant of the matrix corresponding to the system of equations Shows that determinant = 0 and concludes that when $k = 2$ there is no unique solution
(b) M1: A complete method eliminating one variable from the equations using two different pairs of equations. Condone if a different value of k is used A1: Achieves two equations in the same two variables A1: Must give a reason, shows that the equations are a linear multiple of each other therefore they are consistent.
(b) Alternative M1: A complete method eliminating one variable from the equations using two different pairs of equations. Substitutes these equations into one of the original equations in three variables. A1: Achieves two correct equations in two different variables A1: Shows that the equation works therefore they are consistent.
(c) B1: The three planes form a sheaf. They must have full marks in (b) to award this mark.

Q2.

Question	Scheme	Marks	AOs
(a)	a represents the proportion of juvenile chimpanzees that (survive and) remain juvenile chimpanzees the next year.	B1	3.4
		(1)	
(b)(i)	Determinant = $0.82a - 0.08 \times 0.15$	M1	1.1b
	$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1} = \dots \begin{pmatrix} 0.82 & -0.15 \\ -0.08 & a \end{pmatrix}$	M1	1.1b
	$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1} = \frac{1}{0.82a - 0.012} \begin{pmatrix} 0.82 & -0.15 \\ -0.08 & a \end{pmatrix}$	A1	1.1b
		(3)	
(ii)	$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1} \begin{pmatrix} 15360 \\ 43008 \end{pmatrix} = \frac{1}{0.82a - 0.012} \begin{pmatrix} 0.82 \times 15360 - 0.15 \times 43008 \\ (-0.08) \times 15360 + 43008a \end{pmatrix}$ OR forms equations $15360 = aJ_0 + 0.15 \times A_0$ $43008 = 0.08 \times J_0 + 0.82 \times A_0$	M1	3.1a
	$\frac{1}{0.82a - 0.012} [6144 + (43008a - 1228.8)] = 64000$ $\Rightarrow 4915.2 + 43008a = 64000(0.82a - 0.012) \Rightarrow a = \dots$ OR $A_0 = 64000 - J_0 \Rightarrow 43008 = 0.08 \times J_0 + 0.82 \times (64000 - J_0) = J_0 = \dots$ $\Rightarrow a = \frac{15360 - (64000 - J_0)}{J_0} = \dots$	M1	3.1a
	$a = \frac{5683.2}{9472} = 0.60$	A1	1.1b
		(3)	
(iii)	Initial juvenile population = $\frac{"6144"}{"0.48"} = 12800$	M1	3.4
	So change of 2560 juvenile chimpanzees	A1	1.1b
		(2)	

(c)	As the number of juveniles has increased, the model is not initially predicting a decline, so is not suitable in the short term. (Follow through their answer to (b) – but they must have made an attempt at it to find at least a value for J_0)	B1ft	3.5a
		(1)	
(d)	Third category needs to be introduced for chimpanzees aged 40 and above, mature chimpanzees M_n , and a matrix multiplication of increased dimension set up. Accept 3×3 , 3×2 or 2×3 matrices including all three categories in the column vector.	M1	3.5c
	The corresponding matrix model will have the form $\begin{pmatrix} J_{n+1} \\ A_{n+1} \\ M_{n+1} \end{pmatrix} = \begin{pmatrix} a & b & \underline{0} \\ 0.08 & c & 0 \\ 0 & d & e \end{pmatrix} \begin{pmatrix} J_n \\ A_n \\ M_n \end{pmatrix}$ (The underlined zero must be correct but do not be concerned about any values used in the other entries.)	A1	3.3
		(2)	
	(12 marks)		

Notes		
(a)	B1	Correct interpretation. Need not mention survival but must be clear it is the (proportion of) juveniles that remain as juveniles the next year (ie those that survive but don't progress to adulthood). E.g. accept "(number of) juveniles who do not become adults" but do not accept "surviving juveniles".
(b)(i)	M1	Mark part (b) as a whole. Attempts the determinant in terms of a . Allow miscopies for the attempt. Allow $0.82a - 0.12$ as a slip.
	M1	Attempts the form of the inverse, swapped leading diagonals and sign changed on both off diagonals. Allow miscopies of the numbers but the signs must be correct.
	A1	Correct inverse matrix
	(ii)	M1
		Use the inverse matrix and attempts to find the initial juvenile and adult populations. (May have determinant 1 for this mark.) Alternatively, sets up simultaneous equations from the original system, $15360 = aJ_0 + 0.15 \times A_0$ and $43008 = 0.08 \times J_0 + 0.82 \times A_0$. Accept with J_n and A_n or other appropriate variables.
		M1
		Uses the sum of initial populations equals 64000 in an attempt to find a . (May have determinant 1 for this mark.) If using alternative, use of e.g. $A_0 = 64000 - J_0$ in second equation to find J_0 , followed by attempt to find a . Award for an attempt to solve the equations, but don't be too concerned with the algebraic process as long as they are attempting to use all three equations.
		A1
	(iii)	Correct value, $a = 0.6$ (or 0.60 or $\frac{3}{5}$).
		M1
(c)	M1	Uses their a to find the value of J_0 . This mark may be gained for work done in (ii) if the alternative has been used but must have come from a correct method.
		A1
	A1	Correct difference found, as long as there is no contradictory statement – so "decrease of 2560" is A0.
(d)	Blft	Comments that the change is an increase so does not fit the model. Follow through their answer to (b) as long as at least a value for J_0 has been found. If a decrease has been found allow for commenting the model is suitable. If an answer is given to (b)(iii), follow through on whatever their answer is. If no answer has been given, but an initial population found, a comparison should be made between this value and 153600 with conclusion must be consistent with their answer for J_0
(d)	M1	Introduces a third category (may be <i>Mature</i> , <i>Elderly</i> or any suitable letter used) and sets up a matrix multiplication (the left hand side may be missing for this mark) with all three categories in the column vector. The dimension of the matrix should be 3 in at least either row or column, and there should be a 3×1 vector.
	A1	Sets up the new matrix equation, including both sides and making clear the zero (underlined) so that the correct progression that no new juveniles arise from the mature chimpanzees is clear. Overlook other values, though ideally the other two zeroes are shown too, to indicate mature chimpanzees do not regress to adulthood, and juveniles cannot proceed directly to mature chimpanzees.

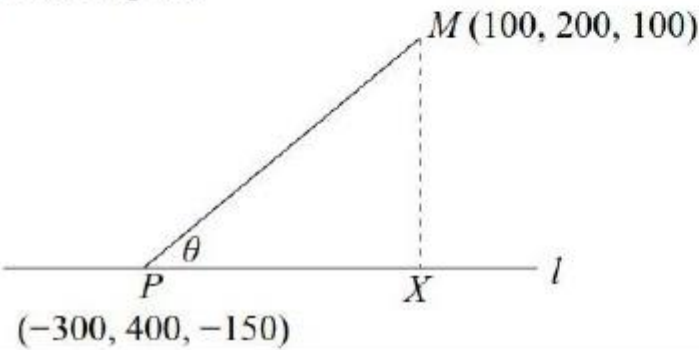
Appendix: Alternatives to (b)

Note that variations may occur with the line equation chosen in part (a), but mark as follows:

Question	Scheme		Marks	AOs
Alt 1 (b)(i)	As per main scheme.		B1 M1	2.2a 3.1b
	$d^2 = (-400 + 600\lambda)^2 + (200 - 100\lambda)^2 + (-250 + 100\lambda)^2$ $= 380000\lambda^2 - 570000\lambda + 262500$ $= 380000\left(\lambda - \frac{3}{4}\right)^2 + 48750 \Rightarrow \lambda = \dots$		dM1	1.1b
	As per main scheme.		M1 A1	3.4 1.1b
(ii)			(5)	
	Length of tunnel is $\sqrt{48750} = \dots$		M1	1.1b
	Awrt 221m from correct working, so completion of square must have been correct. (Must include units)		A1	1.1b
			(2)	
Notes				
(i)	B1M1	As per main scheme.		
	M1	Realises the need to find the distance from the point on the mountain to a general point on the line.		
	dM1	Attempts the distance or distance squared of $\overline{MX}$, expands and completes the square to find the value of λ for which distance is minimum. May obtain other forms for the completed square. Look for $A(B\lambda - C)^2 - D + "262500"$ where $A, B, C, D \neq 0$ but B may be 1.		
	M1A1	As per main scheme.		
(ii)	M1	Correct method for the distance. May be as per main scheme, or via extracting from the completed square constant term.		
	A1	Correct distance, including units. Accept awrt 221 m or $25\sqrt{78}$ m		

Alt 2 (b)(i)	As per main scheme.	B1 M1	2.2a 3.1b
	$d^2 = (-400 + 600\lambda)^2 + (200 - 100\lambda)^2 + (-250 + 100\lambda)^2$ $= 380000\lambda^2 - 570000\lambda + 262500$ $\frac{d}{d\lambda}(d^2) = 0 \Rightarrow 760000\lambda - 570000 = 0 \Rightarrow \lambda = \dots$	dM1	1.1b
	As per main scheme.	M1 A1	3.4 1.1b
(ii)		(5)	
	Length of tunnel is $\sqrt{(150 - 100)^2 + (325 - 200)^2 + (-75 - 100)^2} = \dots$	M1	1.1b
	Awrt 221m from correct working, differentiation etc must have been correct. (Must include units)	A1	1.1b
		(2)	

Notes			
		As per main scheme except for:	
(i)	dM1	Attempts the distance or distance squared of $\overline{MX}$, differentiates and set to zero to find λ for minimum distance.	
(ii)	M1	May substitute λ into the distance squared formula to find distance.	
Alt 3 (b)(i)	$k = 200$	B1	2.2a
	If M is the point on mountain, then e.g (may use Q rather than P)		
	$\overline{MP} = \begin{pmatrix} -400 \\ 200 \\ -250 \end{pmatrix} \Rightarrow \cos \theta = \frac{\begin{pmatrix} -400 \\ 200 \\ -250 \end{pmatrix} \cdot \begin{pmatrix} 600 \\ -100 \\ 100 \end{pmatrix}}{\sqrt{(-400)^2 + 200^2 + (-250)^2} \sqrt{600^2 + (-100)^2 + 100^2}}$ $\Rightarrow \cos \theta = \dots \text{ or } \theta = \dots \text{ (where } \theta \text{ is the angle between the line and } \overline{MP} \text{)}$ $\Rightarrow \overline{PX} = \overline{MP} \cos \theta = \dots$	M1	3.1b
	So e.g.		
	$\overline{OX} = \begin{pmatrix} -300 \\ 400 \\ -150 \end{pmatrix} + \frac{ \overline{PX} }{\left \begin{pmatrix} 600 \\ -100 \\ 100 \end{pmatrix} \right } \begin{pmatrix} 600 \\ -100 \\ 100 \end{pmatrix} = \begin{pmatrix} -300 \\ 400 \\ -150 \end{pmatrix} + \frac{75\sqrt{8}}{100\sqrt{38}} \begin{pmatrix} 600 \\ -100 \\ 100 \end{pmatrix} = \dots$	M1	3.4
	So coordinates of X are $(150, 325, -75)$ Accept as $\begin{pmatrix} 150 \\ 325 \\ -75 \end{pmatrix}$	A1	1.1b
(ii)		(5)	
	Length of tunnel is $ \overline{MP} \sin \theta = \dots$ (oe)	M1	1.1b
	Awrt 221m from correct working. (Must include units)	A1	1.1b
		(2)	

Notes		
(i)	B1	Correct value of k deduced.
	M1	Finds $\overline{MP}$ (or $\overline{MQ}$) and attempts scalar product formula with this and the direction of the line to find the angle or cosine of the angle between line and $\overline{MP}$ (or $\overline{MQ}$)
	dM1	Uses their angle with the cosine to find the length of $\overline{PX}$ (or $\overline{QX}$). Accept equivalent trigonometric methods (e.g. finding opposite side first and using tangent or Pythagoras).
	M1	Uses the length of and $\overline{PX}$ (or $\overline{QX}$) to find the coordinates of the point on the line at shortest distance from M .
	A1	Correct point.
(ii)	A1	Correct method for the distance. May be as per main scheme, or use of sine ratio with their angle between the line and and $\overline{MP}$ (or $\overline{MQ}$). Accept equivalent trigonometric methods.
	A1	Correct distance, including units. Accept awrt 221 m or $25\sqrt{78}$ m
Useful diagram:  $M(100, 200, 100)$ $P(-300, 400, -150)$ X l θ		
Note for P, $\cos \theta = \pm \frac{57}{\sqrt{38}\sqrt{105}}$, $\theta = 25.5\dots^\circ$ and $\overline{PX} = 75\sqrt{38}$ For Q $\cos \theta = \pm \frac{19}{\sqrt{38}\sqrt{29}}$, $\theta = 55.08\dots^\circ$, $\overline{QX} = 25\sqrt{38}$		

Q3.

Question	Scheme	Marks	AOs
	x = value of savings account, y = value of property bond account, z = value of share dealing account. $x + y + z = 5000$ $x + 400 = y$ $0.015x + 0.035y - 0.025z = 79$ or $1.015x + 1.035y + 0.975z = 5079$	M1	3.1b
		A1	1.1b
	Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0.015 & 0.035 & -0.025 \end{pmatrix}$ or $\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1.015 & 1.035 & 0.975 \end{pmatrix}$		
	e.g. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0.015 & 0.035 & -0.025 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5000 \\ -400 \\ 79 \end{pmatrix}$	M1	3.1a
		A1	1.1b
	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0.015 & 0.035 & -0.025 \end{pmatrix}^{-1} \begin{pmatrix} 5000 \\ -400 \\ 79 \end{pmatrix} = \begin{pmatrix} \dots \\ \dots \\ \dots \end{pmatrix}$	M1	1.1b
	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1800 \\ 2200 \\ 1000 \end{pmatrix}$	A1	1.1b
	Tyler invested £1800 in the savings account, £2200 in the property bond account and £1000 in the share dealing account.	A1ft	3.2a
		(7)	
	(7 marks)		

	Question Notes	
	M1	Attempts to set up 3 equations with 3 unknowns.
	A1	At least 2 equations are correct with the appropriate variables defined.
	M1	Sets up a matrix equation of the form, e.g. $\begin{pmatrix} \dots & \dots & \dots \\ \dots & \dots & \dots \\ \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \dots \\ \dots \\ \dots \end{pmatrix},$
		where "... " are numerical values.
	A1	Correct matrix equation (or equivalent).
	M1	Depends on previous M mark. Applies (their A) ⁻¹ $\begin{pmatrix} 5000 \\ \text{their "-400"} \\ \text{their "79"} \end{pmatrix}$ and obtains at
		least one value of x, y or z.
	A1	Correct answer.
	A1ft	Correct follow through answer in context.
	Note	$A^{-1} = \begin{pmatrix} 0.25 & 0.6 & 10 \\ 0.25 & -0.4 & 10 \\ 0.5 & -0.2 & -20 \end{pmatrix} \text{ or } \begin{pmatrix} -9.75 & 0.6 & 10 \\ -9.75 & -0.4 & 10 \\ 20.5 & -0.2 & -20 \end{pmatrix}$